AI Detector report by

QuillBot 2026-04-10 11:45 PM

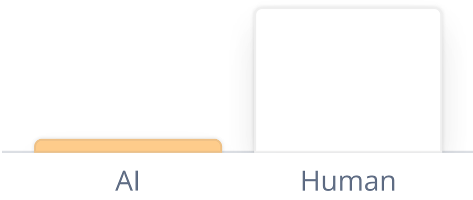
Version 2026-v5.9.1

8,536 Words

Results

9 %

of text is likely AI




| | | |
|---|---|---|
| AI-generated | | 9% |
| Human-written & AI-refined | | 0% |
| Human-written | | 91% |

Caution: Our AI Detector is advanced, but no detectors are 100% reliable, no matter what their accuracy scores claim. Never use AI detection alone to make decisions that could impact a person's career or academic standing.

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Table~\ref{tab:paradigm_compare} further condenses this comparison by grouping methods into paradigms and highlighting the best accuracy achieved within each category.

\begin{table}[h]
\centering
\footnotesize
\caption{Best Accuracy by Learning Paradigm}
\label{tab:paradigm_compare}
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\section{Proposed Methodology}

The full pipeline is illustrated in Figure~\ref{fig:pipeline}. At a high level, it operates in two stages: an unsupervised pretraining phase where the CNN-LSTM encoder learns signal representations through contrastive loss on unlabeled windows, followed by a supervised fine-tuning phase where a lightweight classification head is trained on top of the frozen (or partially unfrozen) encoder using labeled stress data.

\begin{figure}[h]

\centering

\includegraphics[width=\linewidth]{ssl_pipeline.png}

\caption{Overview of the proposed self-supervised learning pipeline. The model learns representations from unlabeled data using contrastive learning before fine-tuning for stress classification.}
\label{fig:pipeline}
\end{figure}

\subsection{Overview of the Framework}

The pipeline separates the problem into two concerns: learning what physiological signals look like in general (pretraining) and learning which patterns correspond to stress specifically (fine-tuning). [redacted]

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\begin{figure}[h]
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\includegraphics[width=\linewidth]{model_architecture.png}
\caption{Proposed model architecture combining CNN and LSTM layers for feature extraction and stress classification.}
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& Dataset & Year & Description & Pros & Cons \\
\hline
1 & WESAD~\cite{b10} & 2018 & Multimodal wearable stress dataset & Real-world signals & Limited subjects \\
\hline
2 & SWELL-KW~\cite{b33} & 2013 & Workplace stress dataset & Controlled environment & Less diversity \\
\hline
3 & DEAP~\cite{b34} & 2012 & Emotion dataset with EEG & Rich signals & Not stress-specific \\
\hline
4 & AMIGOS~\cite{b35} & 2018 & Multimodal emotion dataset & Diverse features & Complex preprocessing \\
\hline
5 & Proposed Use & 2024 & SSL-based physiological dataset & Uses unlabeled data & Requires pretraining \\
\hline
\end{tabular}

\end{table*}

Table~\ref{tab:dataset} compares several publicly available stress and affect datasets. Each has different strengths: DEAP \cite{b34} and AMIGOS \cite{b35} offer rich EEG-based emotional signals but are not explicitly stress-focused; SWELL-KW \cite{b33} provides naturalistic workplace recordings but limited physiological diversity. We chose WESAD \cite{b10} specifically because it includes the three wrist-worn modalities central to our pipeline (EDA, BVP, TEMP) alongside standardized stress induction, making it the closest available approximation to a real deployment scenario.

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[redacted] Figure~\ref{fig:accuracy} summarizes accuracy across all five systems.

\begin{figure}[h]
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\includegraphics[width=\linewidth]{accuracy.png}
\caption{Accuracy comparison of different models, showing superior performance of the proposed SSL-based approach.}

\label{fig:accuracy}
\end{figure}

The SSL model reaches 91.5\% accuracy, which is +4.6 points above the LSTM baseline and +13.0 points above SVM. [redacted]

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\begin{table}[h]
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\caption{Performance Comparison of Different Models}
\begin{tabular}{|c|c|c|c|c|}
\hline
Model & Accuracy & Precision & Recall & F1-Score \\
\hline
SVM & 78.5\% & 76.2\% & 74.8\% & 75.5\% \\
Random Forest & 81.3\% & 80.1\% & 79.5\% & 79.8\% \\
CNN & 84.7\% & 83.5\% & 84.0\% & 83.7\% \\
LSTM & 86.9\% & 85.8\% & 86.2\% & 86.0\% \\
Proposed SSL Model & \textbf{91.5\%} & \textbf{90.8\%} & \textbf{91.2\%} & \textbf{91.0\%} \\
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\end{tabular}
\end{table}

The results clearly indicate that the proposed SSL-based model outperforms all baseline methods across all evaluation metrics.

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[redacted] The SSL initialization led to noticeably faster convergence during fine-tuning — the classification head reached near-optimal accuracy within the first 10 epochs, compared to 20+ for the LSTM trained from scratch. [redacted]

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\section{Ethical Implications and Responsible AI}

Deploying a stress detection system outside a research lab raises ethical questions that deserve more than a footnote. Physiological data is among the most sensitive information a person generates — it can reveal anxiety disorders, medication use, and emotional states that the individual may not have chosen to disclose. [redacted]

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[redacted] His research interests sit at the intersection of machine learning and physiological sensing, with a particular focus on self-supervised representation learning and cross-subject generalization in health monitoring systems. [redacted]

[redacted] He holds an impressive academic background, boasting qualifications such as B.E., M.Tech., and Ph.D. in Computer Science and Engineering. His expertise lies in Optimization, MANET, Network Security, and Machine Learning. As an accomplished author, Dr. Bairwa has published eight books and is actively involved in research across various areas of his interest. He is a valued member of prestigious organizations, including ISTE, ACM, and IEEE, which reflects his commitment to professional growth and networking. In addition to his academic pursuits, Dr. [redacted] Bairwa has significantly contributed to the academic ecosystem during his tenure. He has taken initiatives to enhance the learning environment within the classroom, developed cutting-edge academic and professional programs, and adapted them to meet the evolving needs of the industry in today's dynamic business landscape. [redacted] Amit Kumar Bairwa has collaborated with fellow academicians and played an integral role in setting and maintaining high curriculum standards. He actively contributes to developing mission statements and establishes performance goals and objectives, ensuring that the educational institution consistently excels in its academic pursuits. [redacted]